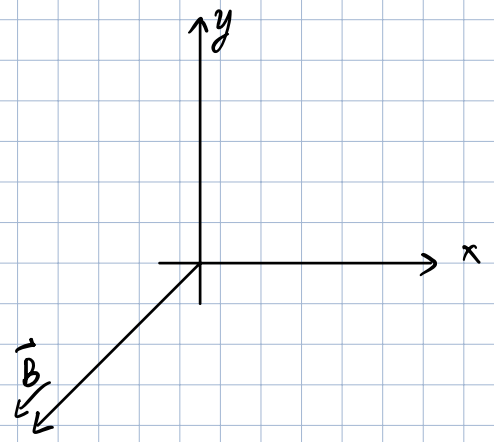
2.2 非均匀磁场下漂移.

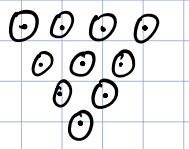


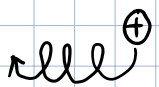
① $\nabla \vec{B} \perp \vec{B}$

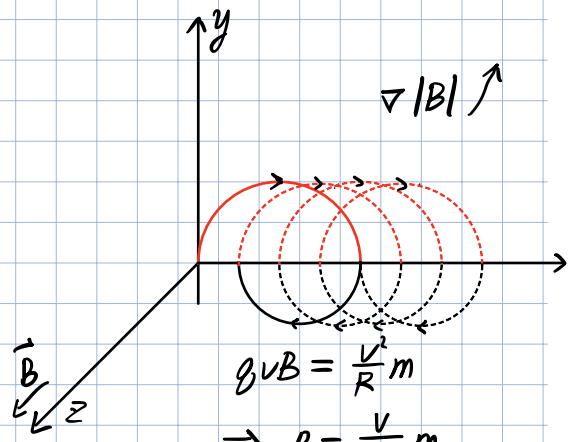
② 弯曲 B

③ 磁镜. $\nabla B \parallel B$.

2.2.1 $\nabla B \perp \vec{B}$ B的梯度漂移.







$qvB = \frac{v^2}{R}m$

$\Rightarrow R = \frac{v}{Bq}m$.

$\vec{F} = q\vec{v}\times\vec{B}$ 对回转一周的平均. $\bar{F}_x = 0$ B↑. R↓.

$\bar{F}_y$?

未扰动轨道法.

粒子向上、向下所用时间相同, ∴ 用粒子未扰动轨道平均求 $\bar{F}_y$.

未扰动轨道: $v_x = Re(v_\perp e^{i\omega_c t}) = v_\perp \cos\omega_c t$

$F_y = q v_x B_z(y)$

Taylor 展开.

$\vec{B} = \vec{B}_0 + (\vec{r}\cdot\nabla)\vec{B} + \cdots$, $B_z = B_0 + y\frac{\partial}{\partial y}B_z + \cdots$

展开要求:

$\frac{r_L}{L} \ll 1$

L: 标度长.

$$B_z = B_0 + r_\perp \cos\omega_c t \frac{\partial}{\partial y} B_z$$

$$\vec{F_y} = -q v_x B_z(y) = -q v_\perp \cos(\omega_c t)\left[B_0 \pm r_\perp (\cos\omega_c t)\frac{\partial B_z}{\partial y}\right]$$

积分 $\overline{F_y} = \mp q v_\perp r_L \frac{1}{2}\frac{\partial B_z}{\partial y}$

$$\vec{v_{gc}} = \frac{1}{q}\frac{\vec{F_y}}{|B|}\hat{x} = \mp \frac{v_\perp r_L}{B}\frac{1}{2}\frac{\partial B_z}{\partial y}\hat{x}.$$

$\frac{\partial B_z}{\partial y} \rightarrow \nabla B$.

$$\vec{v_{gc}} = \pm\frac{1}{2} v_\perp r_L \frac{\vec{B}\times(\nabla B)}{B^2}$$

静电流密度 $\vec{J} = q\vec{v} = ne\vec{v_{gc}}$

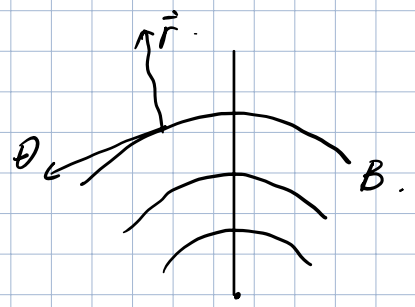
considering 准中性条件： $n_{电} \approx n_{离}$

$$\vec{J} = n(M_{ion} + m_{e^-})\frac{\vec{q}\times\vec{B}}{B^2} \quad (\vec{F} = m\vec{g}).$$

电流产生磁场.

2.2.2 弯曲B场：曲率漂移.

1. 曲率漂移.



$|B_0| = const.$

离心力 $F_{cf} = \frac{mv_{//}^2}{R_c}\hat{r} = mv_{//}^2\frac{\hat{R_c}}{|R_c|^2}$.

曲率漂移：$\vec{v_R} = \frac{1}{\varphi}\frac{\vec{F_\varphi}\times\vec{B}}{|B|^2} = \frac{mv_{//}^2}{q|B|^2}\frac{\vec{R_c}\times\vec{B}}{R_c^2}$

2、总漂移. 曲率 + 梯度.

$$v = \frac{m}{q}\frac{\vec{R_c}\times\vec{B}}{R_c^2B^2}\left(v_{//}^2 + \frac{1}{2}v_\perp^2\right).$$

2分量

$$K_B T_e = \frac{1}{2}mv_{th}^2.$$

$$\frac{2K_BT_e}{m}.$$

$$v = \frac{m}{qB}\frac{1}{k_c}\frac{\vec{R_c}\times\vec{B}}{R_c\cdot B}\left(2\frac{k_BT}{m}\right). = \sim.$$

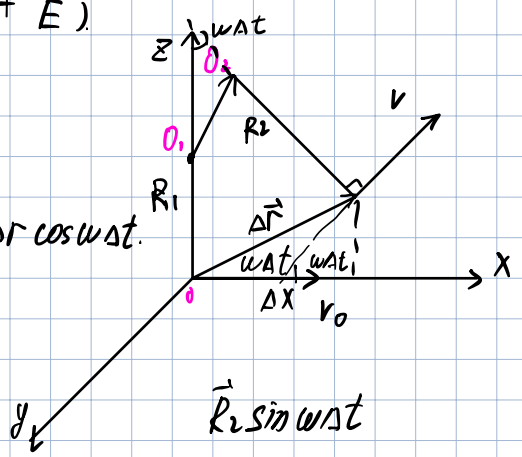
$$= \frac{1}{\omega_c}\frac{1}{R_c}v_{th}^2\hat{y}$$

$$m\frac{d\vec{v}}{dt} = q(\vec{v}\times\vec{B} + \vec{E})$$

$$E\,\Delta x = \frac{1}{2}m\Delta v^2$$

$$= E\Delta r\cos\omega t = E\Delta r\cos\omega\Delta t.$$

$$-\vec{R_1} + \Delta\vec{r} + \vec{R_2}$$



$$\vec{R_2}\sin\omega\Delta t$$

$$\frac{\frac{1}{2}m\Delta v^2}{E}$$

$$\frac{v+\Delta v}{Bq}$$

$$\frac{v^2}{R} = Bvq = \omega^2 R$$

$$= v\omega$$

$$\omega = Bq$$

